

DLOPs Assignment-1

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Link for the colab notebook :

https://colab.research.google.com/drive/1_GaXraX9kE4Gjm_7j1wQFsINd6gv12de?usp=sharing

Performance and Compute Analysis of ResNet-18 / ResNet-50 and SVM on MNIST & FashionMNIST

Problem statement

Q1(a) Train deep learning models (ResNet-18 and ResNet-50; pretrained=False) on MNIST and FashionMNIST with a target 70% / 10% / 20% train/val/test split. Vary hyperparameters (batch size, optimizer, learning rate) and epochs; keep AMP on. Report test classification accuracy.

Q1(b) Train SVM classifiers on MNIST and FashionMNIST with multiple kernels (poly, rbf) and SVM hyperparameter settings; report test accuracy and training time (ms).

Q2 Compare training performance for FashionMNIST when running models on CPU vs GPU: provide test accuracy, training time, and FLOPs/parameter counts for ResNet models.

Datasets and preprocessing

Datasets

- MNIST: 70k images (60k train, 10k test), 10 classes of handwritten digits.
- FashionMNIST: 70k images of fashion items (clothing), 10 classes; same image size & counts as MNIST.

Splits used in this study

- Intended: 70% (49,000) train, 10% (7,000) val, 20% (14,000) test.

- Implementation note: used 49,000 train and 7,000 validation from the original training set (60,000), and used the official 10,000-sample test set as the test partition.

Preprocessing

- For SVM experiments:
 - Convert images to flattened vectors (784 dims).
 - Standardize features (zero mean, unit variance) using StandardScaler.
 - Optional PCA to reduce to `n_components=50` (reduces training time and memory for SVM), with reported explained variance.
- For ResNet experiments:
 - Convert grayscale → 3-channel using `transforms.Grayscale(num_output_channels=3)` or adapt conv1 to accept 1 channel (both approaches appear in your code: you used 3-channel + resize for ResNet training; for ptflops/FashionMNIST you adapted conv1 to 1 channel).
 - Resize to 224×224 before feeding into standard ResNet (in training code) and normalize with ImageNet mean/std.
 - Data loaders with `pin_memory=True` and `num_workers=2`.

Random seeds

- `torch.manual_seed(42)` and `np.random.seed(42)` for reproducibility across runs.

Q1(b) : SVM experiments (present in the start of the colab codebase)

- Preprocessing: flatten, standardize, PCA to 50 components (by default; state whether you ran with `use_pca=True`).
- Hyperparameter grid :

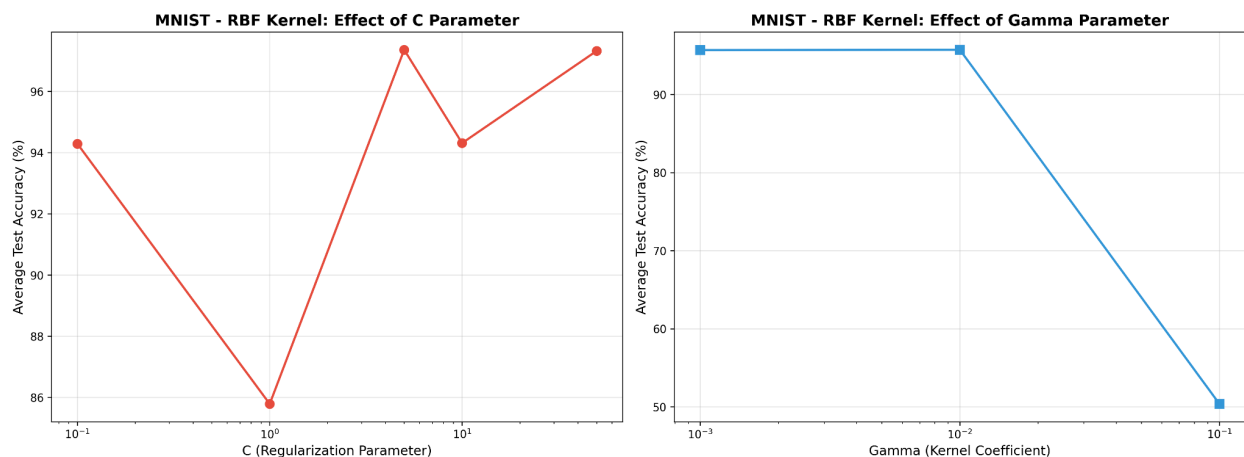
- RBF: $C \in \{0.1, 1.0, 10.0, 5.0, 50.0\}$; $\gamma \in \{\text{scale, auto, numeric values like } 0.001, 0.01, 0.1\}$.
- Poly: $\text{degree} \in \{2, 3, 4\}$, $\text{coef0} \in \{0.0, 1.0\}$, C values as above.
- Metrics recorded: training time (ms), train/val/test accuracy, weighted F1, precision, recall.

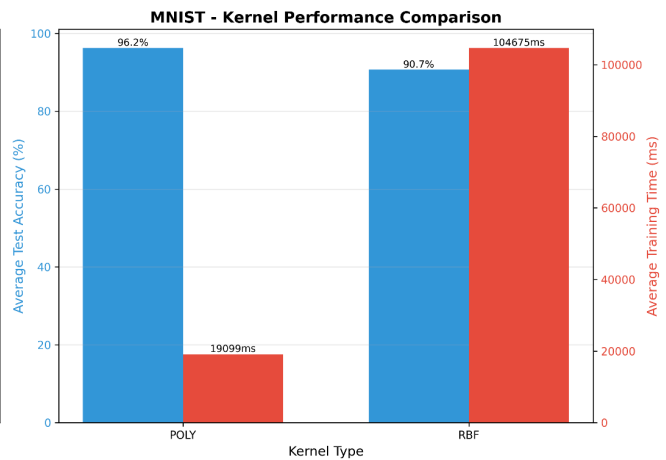
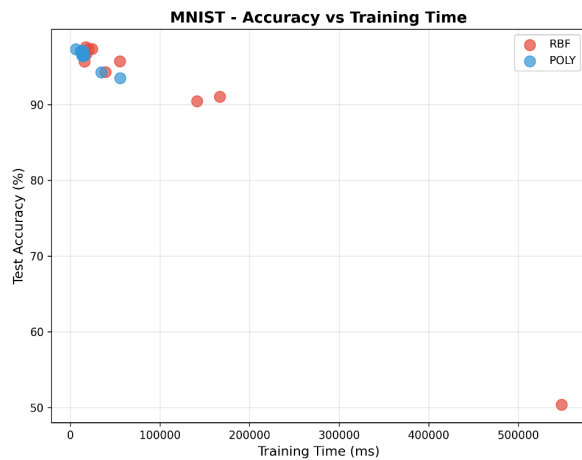
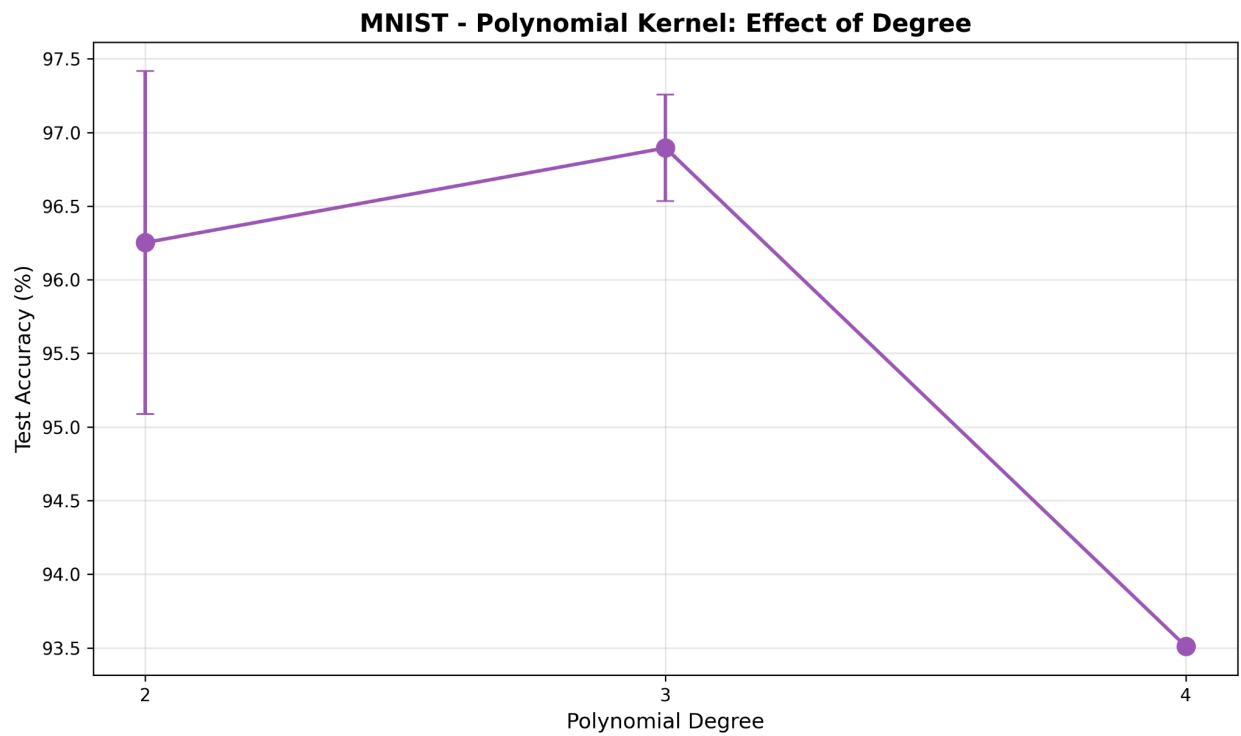
Results on the SVM analysis for both the datasets :

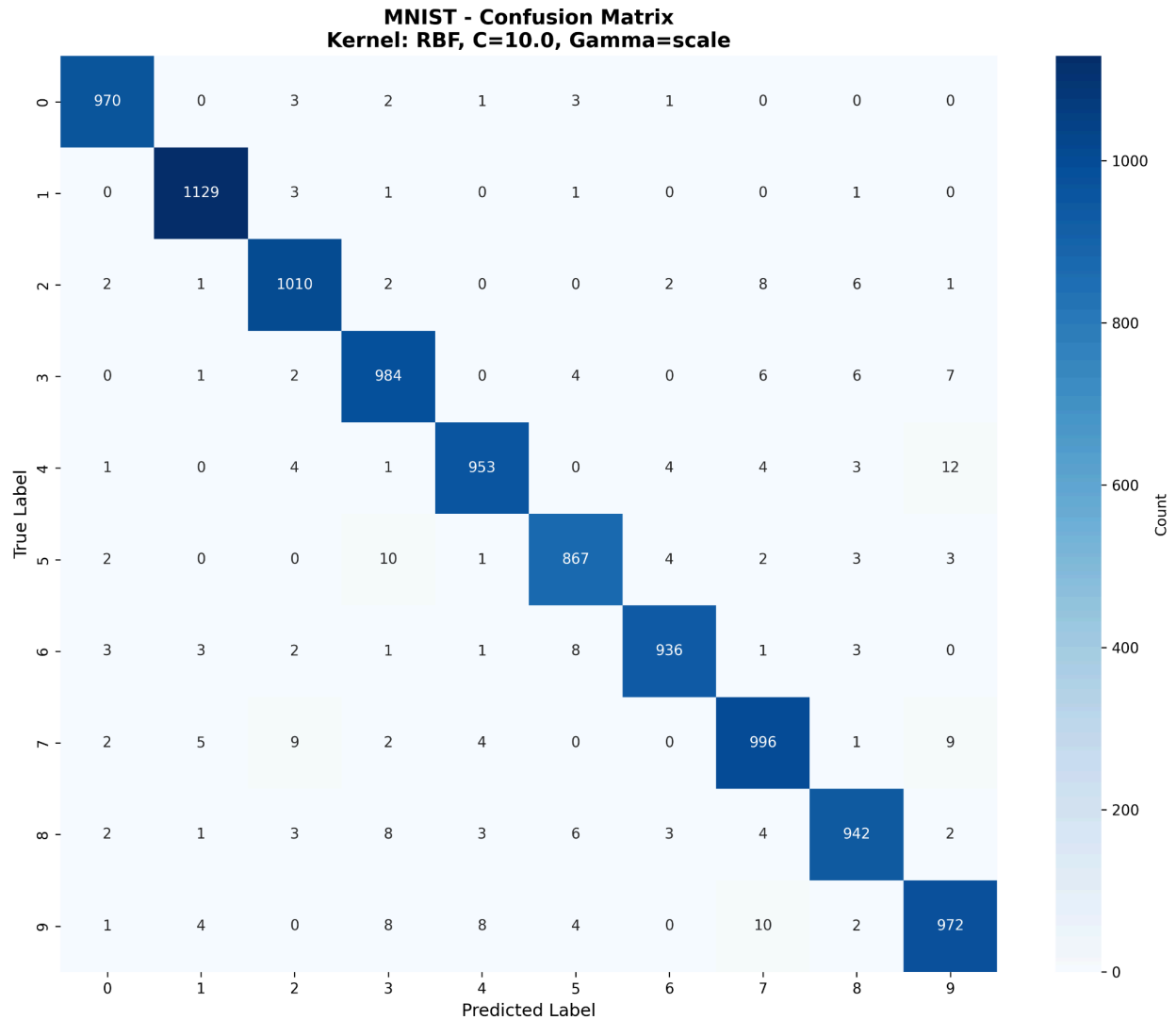
A	B	C	D	E	F	G	H	I	J	K	L	M	N
Dataset	Experiment	Kernel	C	Gamma	Degree	Coef0	Training Time (m)	Train Accuracy (%)	Val Accuracy (%)	Test Accuracy (%)	Test F1-Score (%)	Test Precision (%)	Test Recall (%)
MNIST	1	rbf	0.1	scale	3	0	39270.32	94.7	94.3	94.29	94.28	94.29	94.29
MNIST	2	rbf	1	scale	3	0	17956.4	98.08	96.9	96.7	96.7	96.7	96.7
MNIST	3	rbf	10	scale	3	0	17228.68	99.58	97.79	97.59	97.59	97.59	97.59
MNIST	4	rbf	1	auto	3	0	141499.44	99.82	90.83	90.46	91.22	93.53	90.46
MNIST	5	rbf	10	auto	3	0	166879.5	100	91.43	91.03	91.7	93.74	91.03
MNIST	6	rbf	1	0.001	3	0	15772.77	96.41	95.94	95.68	95.67	95.68	95.68
MNIST	7	rbf	1	0.01	3	0	55265.3	99.51	96.03	95.72	95.78	96.01	95.72
MNIST	8	rbf	1	0.1	3	0	548257.36	100	51.31	50.37	56.03	90.29	50.37
MNIST	9	poly	0.1	scale	2	0	34436.67	94.87	94.24	94.25	94.27	94.33	94.25
MNIST	10	poly	1	scale	2	0	14677.43	97.61	96.39	96.41	96.41	96.42	96.41
MNIST	11	poly	10	scale	2	0	12852.21	99.11	97.09	97.12	97.12	97.12	97.12
MNIST	12	poly	1	scale	3	0	15448.06	97.68	96.26	96.47	96.47	96.5	96.47
MNIST	13	poly	1	scale	4	0	55683.13	95.61	93.5	93.51	93.66	94.18	93.51
MNIST	14	poly	1	scale	3	1	6029.25	98.91	97.4	97.3	97.3	97.3	97.3
MNIST	15	poly	1	auto	2	0	13048.5	99.77	96.91	96.45	96.45	96.46	96.45
MNIST	16	poly	1	auto	3	0	13121.57	99.99	97.04	96.75	96.75	96.76	96.75
MNIST	17	rbf	5	scale	3	0	24205.86	99.27	97.7	97.36	97.36	97.36	97.36
MNIST	18	rbf	50	scale	3	0	20413.27	99.93	97.49	97.32	97.32	97.32	97.32
MNIST	19	poly	5	scale	2	0	11338.22	98.73	97.09	97.03	97.03	97.03	97.03
MNIST	20	poly	5	scale	3	0	14350.04	99.05	97.1	97.06	97.06	97.07	97.06

The above table shows the results for the MNIST dataset on the SVM task .

The various analysis performed for this experiment are as follows in plots :

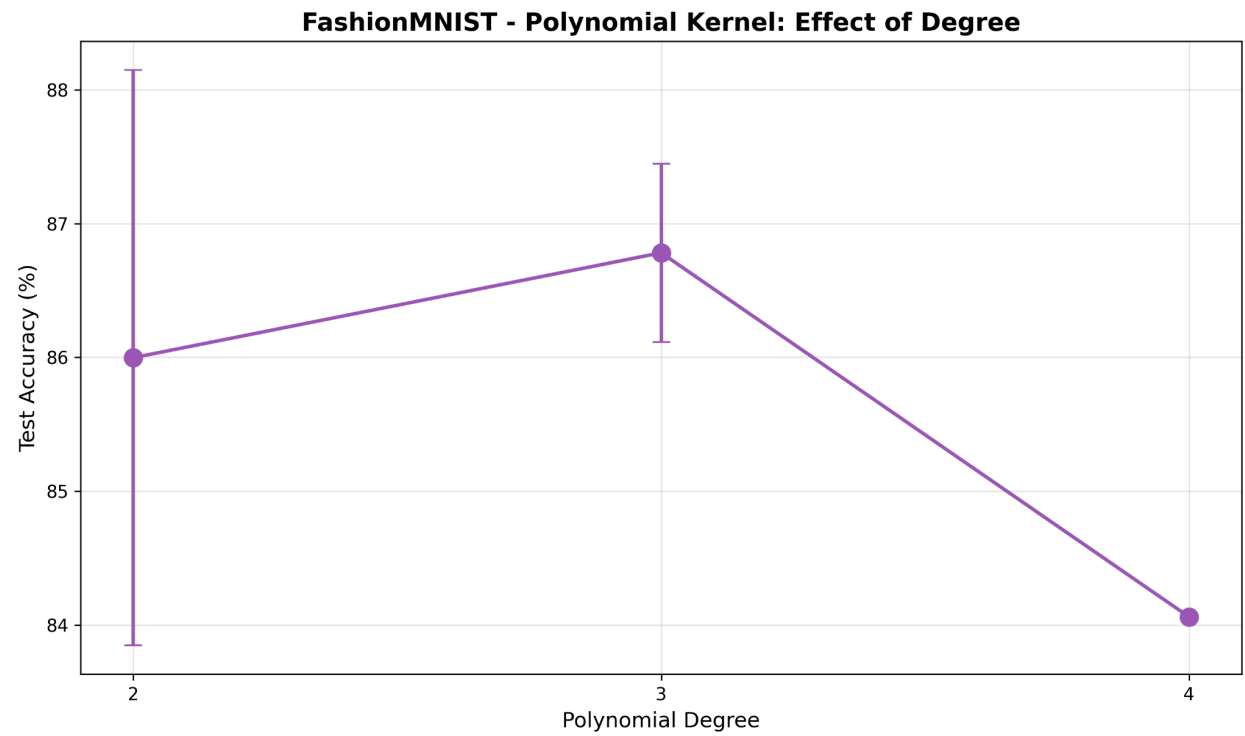
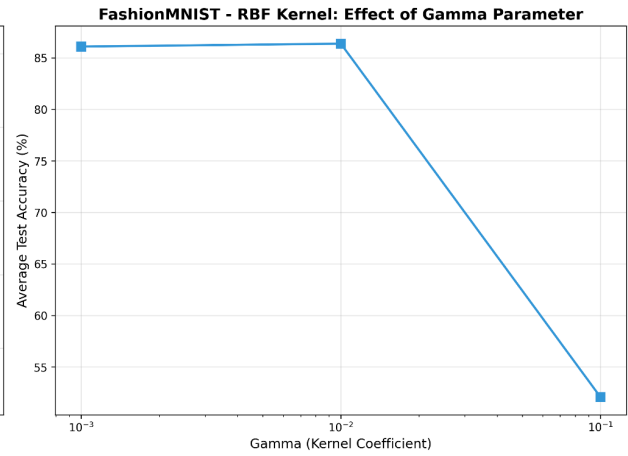
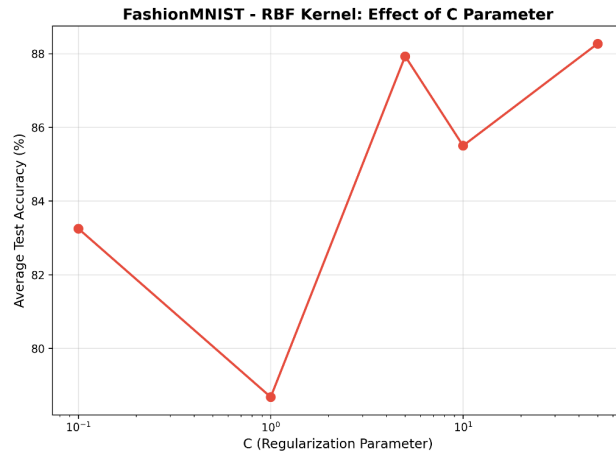


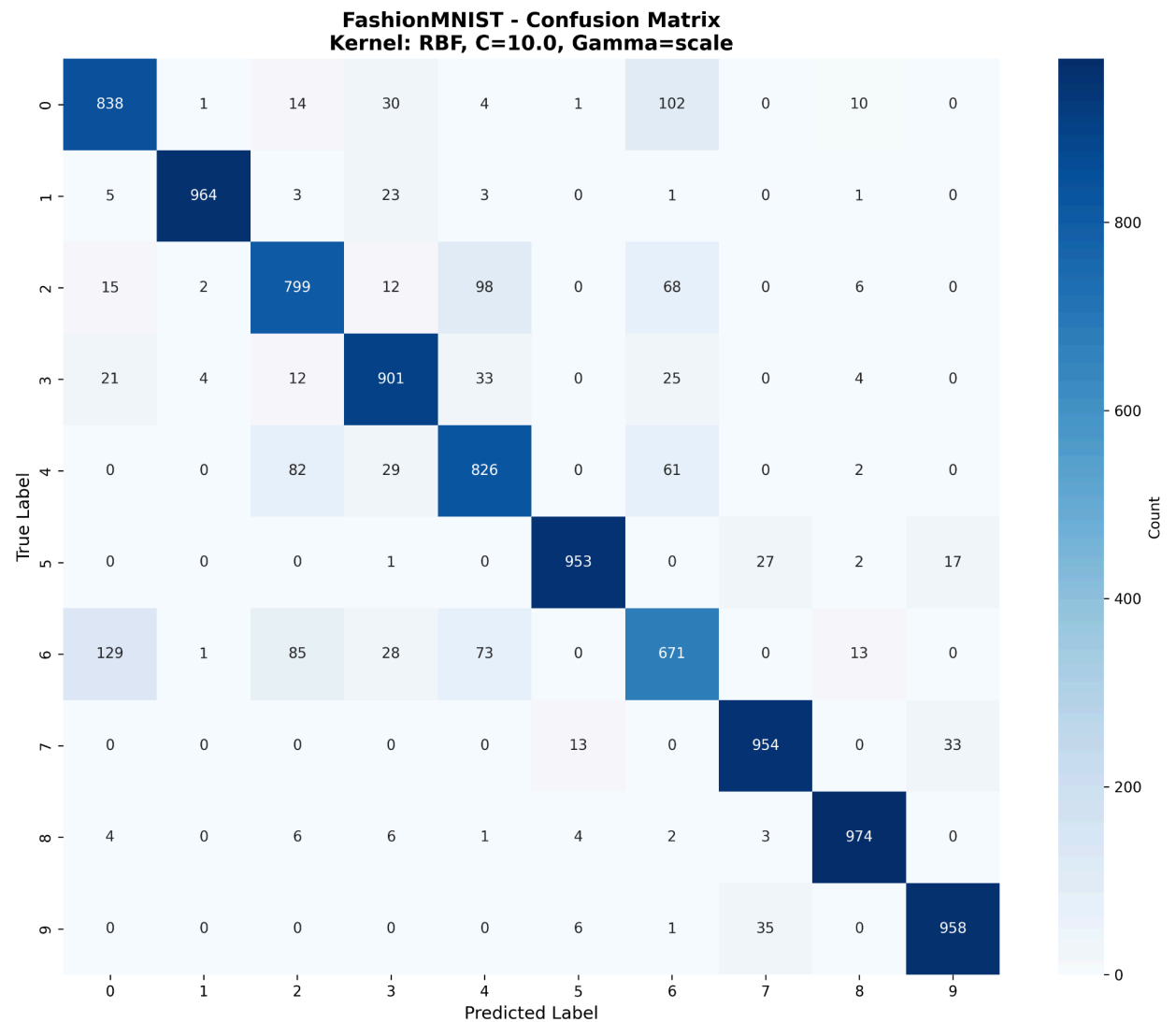
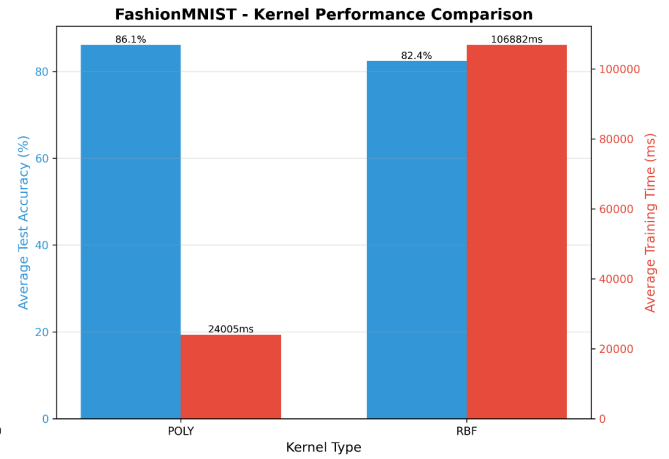
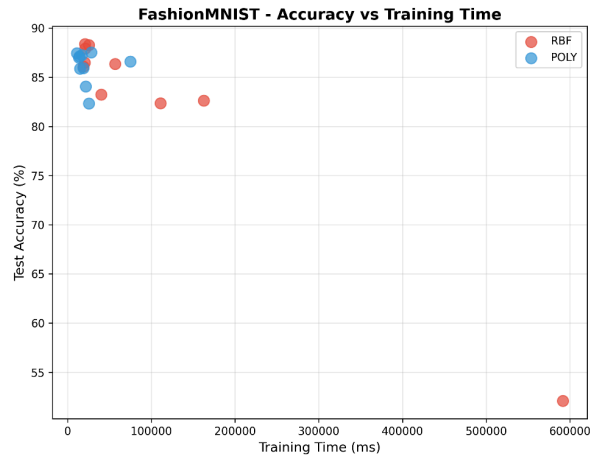




The result for the SVM task on the fashionMNIST dataset are as follows :

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Dataset	Experiment ID	Kernel	C	Gamma	Degree	Coef0	Training Time (n	Train Accuracy	Val Accuracy (%)	Test Accuracy (%)	Test F1-Score (%)	Test Precision (%)	Test Recall (%)
FashionMNIST	1	rbf		0.1 scale	3	0	40318.28	84.33	84.37	83.25	83.06	83.04	83.25
FashionMNIST	2	rbf		1 scale	3	0	20607.67	88.81	87.76	86.5	86.43	86.43	86.5
FashionMNIST	3	rbf		10 scale	3	0	20697.87	92.77	89.94	88.38	88.34	88.34	88.38
FashionMNIST	4	rbf		1 auto	3	0	111096.08	97.08	83.64	82.35	82.6	84.26	82.35
FashionMNIST	5	rbf		10 auto	3	0	162821.28	99.96	83.81	82.62	82.86	84.29	82.62
FashionMNIST	6	rbf		1	0.001	3	0	18804.67	87.69	86.09	86.03	86.03	86.09
FashionMNIST	7	rbf		1	0.01	3	0	56678.82	94.37	87.54	86.37	86.25	86.35
FashionMNIST	8	rbf		1	0.1	3	0	591494.4	99.95	53.13	52.08	57.89	83.07
FashionMNIST	9	poly		0.1 scale	2	0	25397.19	83.29	83.27	82.32	82.5	82.83	82.32
FashionMNIST	10	poly		1 scale	2	0	14733.89	87.89	87.33	85.87	85.87	85.9	85.87
FashionMNIST	11	poly		10 scale	2	0	17145.27	90.55	89.13	87.27	87.25	87.26	87.27
FashionMNIST	12	poly		1 scale	3	0	18938.91	88.33	87.59	85.93	86.02	86.15	85.93
FashionMNIST	13	poly		1 scale	4	0	21712.76	87.24	85.67	84.06	84.46	85.32	84.06
FashionMNIST	14	poly		1 scale	3	1	10956.52	90.64	89.31	87.45	87.42	87.43	87.45
FashionMNIST	15	poly		1 auto	2	0	28424.69	93.21	89.01	87.53	87.5	87.52	87.53
FashionMNIST	16	poly		1 auto	3	0	74999.97	99.29	87.73	86.61	86.6	86.65	86.61
FashionMNIST	17	rbf		5 scale	3	0	20945.39	91.55	89.59	87.93	87.89	87.89	87.93
FashionMNIST	18	rbf		50 scale	3	0	25354.21	95.62	89.94	88.27	88.23	88.22	88.27
FashionMNIST	19	poly		5 scale	2	0	13654.3	89.84	88.9	87	86.98	86.99	87
FashionMNIST	20	poly		5 scale	3	0	14091.01	91.19	89.06	87.14	87.18	87.24	87.14





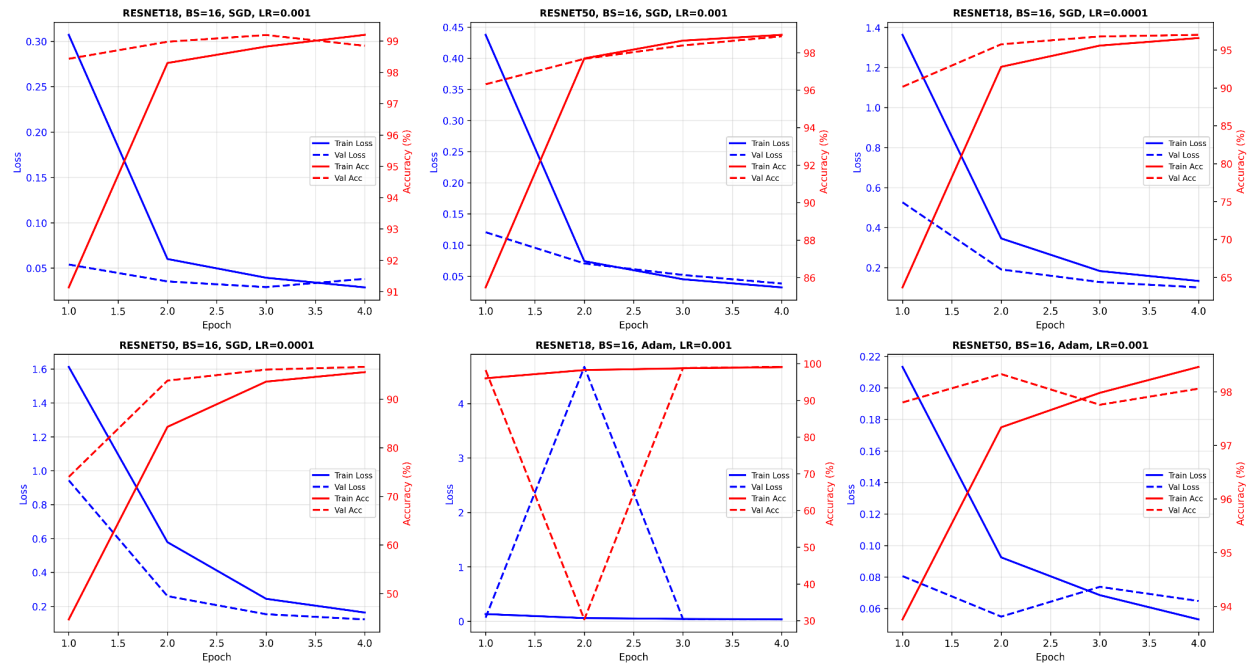
Q1(a) : ResNet experiments (present after Q1b in the colab notebook)

- Images converted/resized for ResNet and normalized with ImageNet stats.
- Experiments run for both models over all combinations in Config.EXPERIMENTS × Config.EPOCHS_LIST.
- Loss: CrossEntropyLoss(); scheduler ReduceLROnPlateau monitors validation accuracy.
- Mixed precision training (torch.cuda.amp + GradScaler) used to reduce memory and possibly speed up GPU training.
- Early model save: best validation accuracy with torch.save('best_model.pth'), then load for final test evaluation.

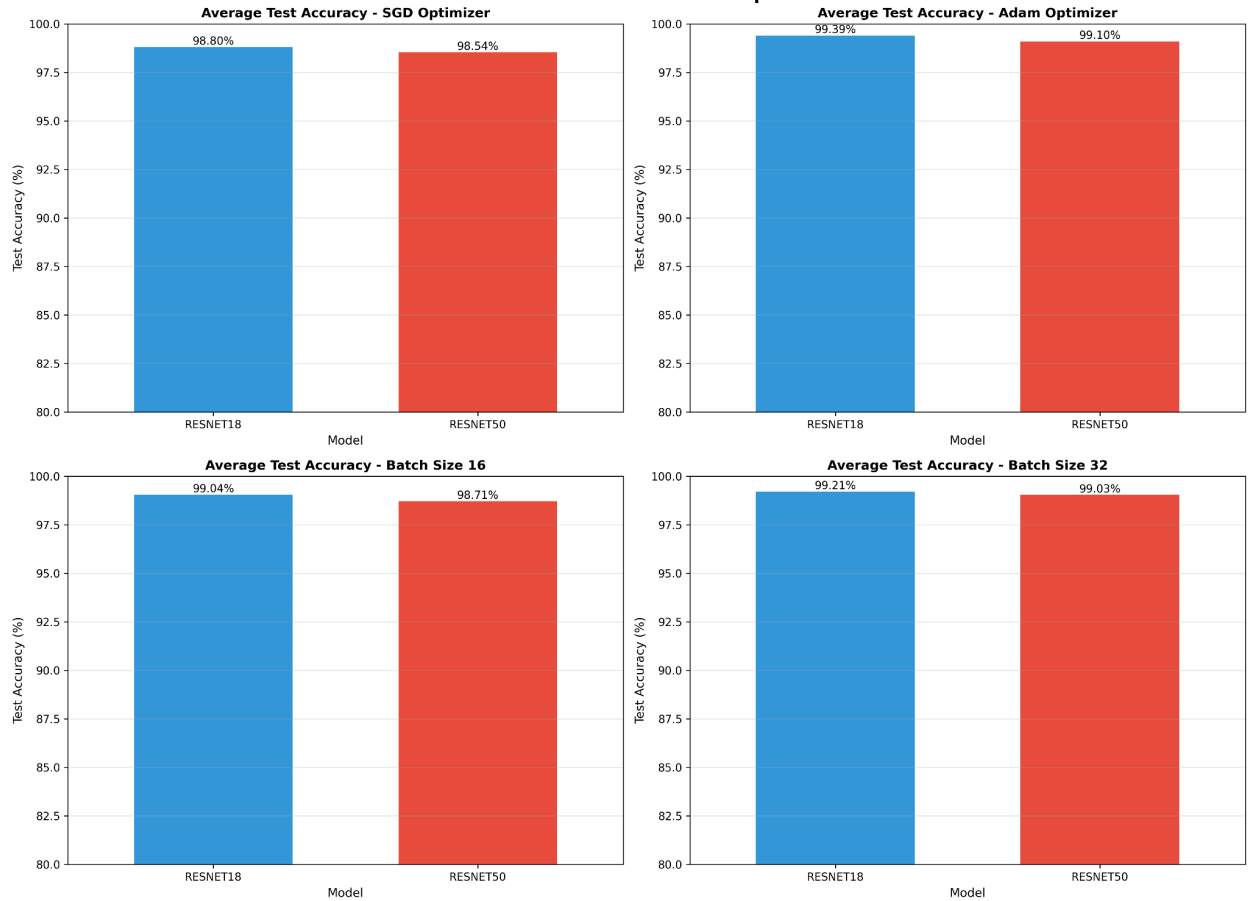
Dataset	Model	Batch Size	Optimizer	Learning Rate	Epochs	Best Val Accuracy	Test Accuracy
MNIST	RESNET18	16	SGD	0.001	4	99.19	99.13
MNIST	RESNET50	16	SGD	0.001	4	98.87	99.02
MNIST	RESNET18	16	SGD	0.0001	4	97	97.43
MNIST	RESNET50	16	SGD	0.0001	4	96.6	97.08
MNIST	RESNET18	16	Adam	0.001	4	99.11	99.18
MNIST	RESNET50	16	Adam	0.001	4	98.33	98.4
MNIST	RESNET18	16	Adam	0.0001	4	99.19	99.38
MNIST	RESNET50	16	Adam	0.0001	4	98.99	99.15
MNIST	RESNET18	32	SGD	0.001	4	98.77	98.78
MNIST	RESNET50	32	SGD	0.001	4	98.76	98.84
MNIST	RESNET18	32	Adam	0.0001	4	99.1	99.3
MNIST	RESNET50	32	Adam	0.0001	4	98.9	99.11
MNIST	RESNET18	16	SGD	0.001	8	99.47	99.39
MNIST	RESNET50	16	SGD	0.001	8	99.23	99.01
MNIST	RESNET18	16	SGD	0.0001	8	98.37	98.78
MNIST	RESNET50	16	SGD	0.0001	8	98.16	98.27
MNIST	RESNET18	16	Adam	0.001	8	99.39	99.51
MNIST	RESNET50	16	Adam	0.001	8	99.26	99.4
MNIST	RESNET18	16	Adam	0.0001	8	99.43	99.53
MNIST	RESNET50	16	Adam	0.0001	8	99.21	99.36
MNIST	RESNET18	32	SGD	0.001	8	99.29	99.28
MNIST	RESNET50	32	SGD	0.001	8	98.9	99.01
MNIST	RESNET18	32	Adam	0.0001	8	99.33	99.47
MNIST	RESNET50	32	Adam	0.0001	8	99.2	99.18

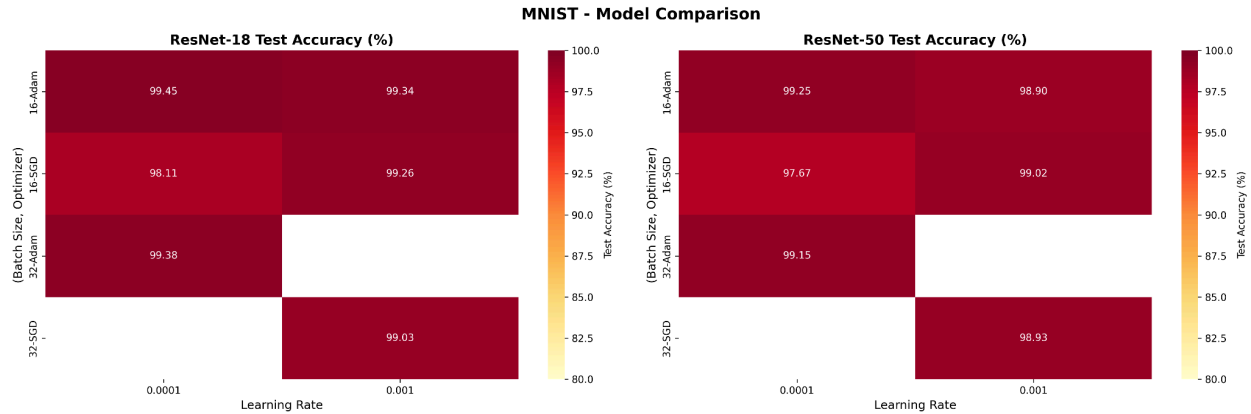
The above are the results for the experimentations on the MNIST dataset .

MNIST - Training Curves



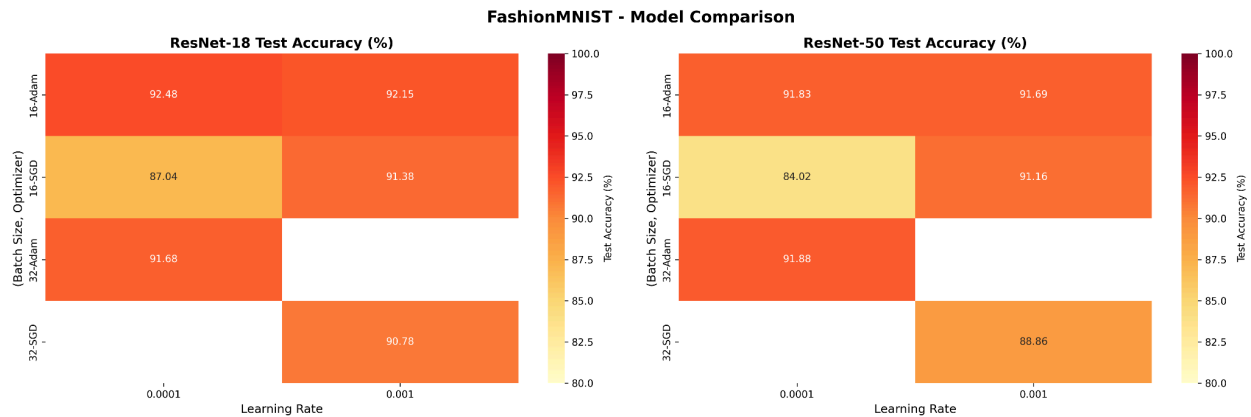
MNIST - Model Performance Comparison



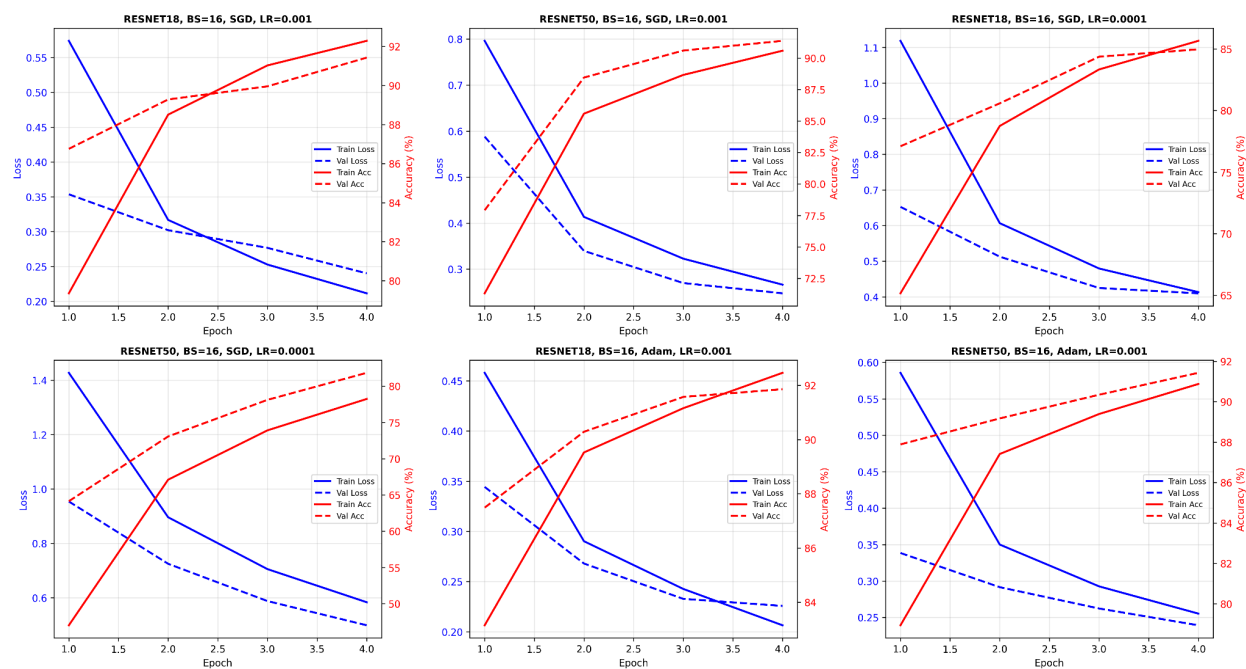


And for the fashionMNIST dataset , the results are as follows :

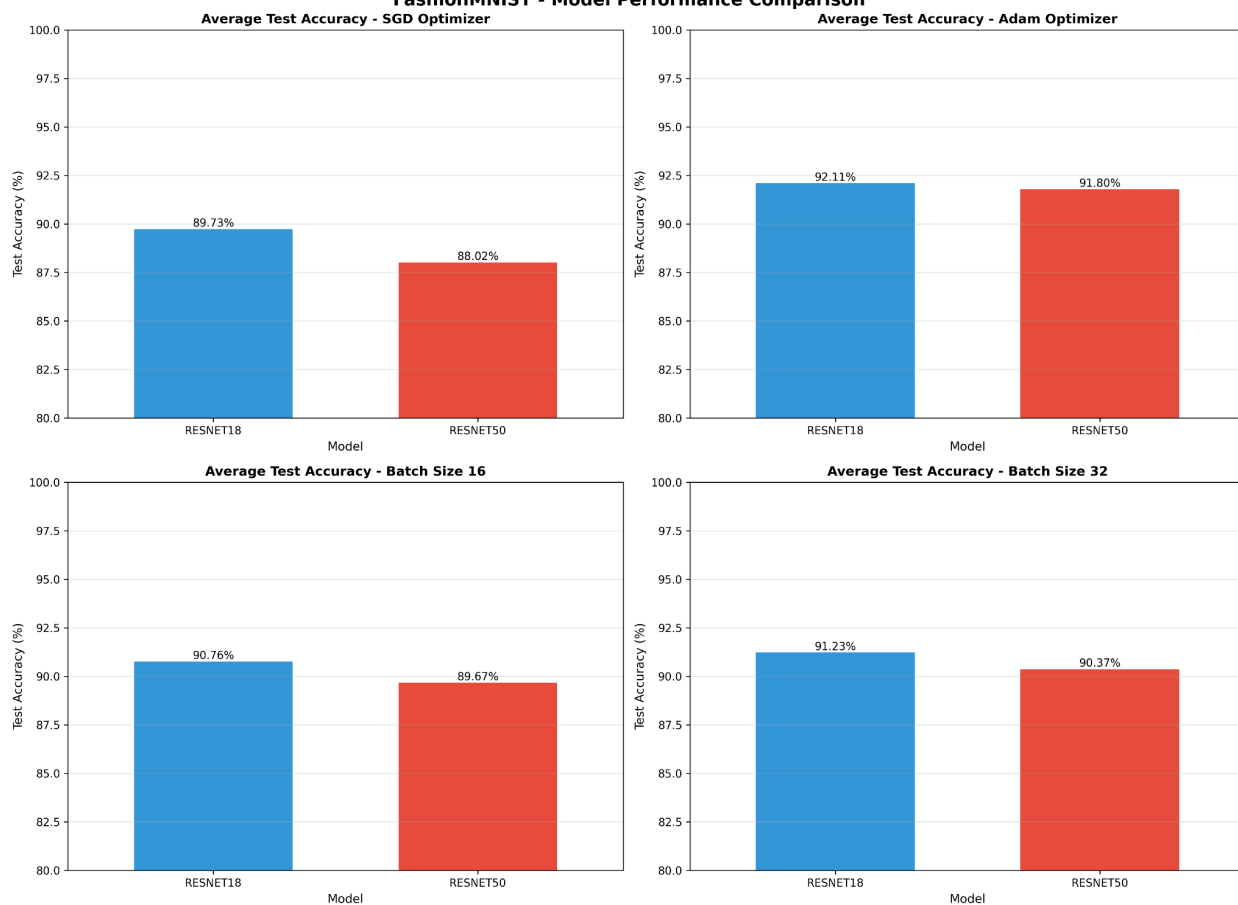
Dataset	Model	Batch Size	Optimizer	Learning Rate	Epochs	Best Val Accuracy	Test Accuracy
FashionMNIST	RESNET18	16	SGD	0.001	4	91.43	90.92
FashionMNIST	RESNET50	16	SGD	0.001	4	91.36	90.66
FashionMNIST	RESNET18	16	SGD	0.0001	4	84.97	84.38
FashionMNIST	RESNET50	16	SGD	0.0001	4	81.83	81.16
FashionMNIST	RESNET18	16	Adam	0.001	4	91.86	91.33
FashionMNIST	RESNET50	16	Adam	0.001	4	91.43	90.96
FashionMNIST	RESNET18	16	Adam	0.0001	4	93.01	92.27
FashionMNIST	RESNET50	16	Adam	0.0001	4	91.76	91.14
FashionMNIST	RESNET18	32	SGD	0.001	4	91.07	90.3
FashionMNIST	RESNET50	32	SGD	0.001	4	88.67	88.12
FashionMNIST	RESNET18	32	Adam	0.0001	4	91.57	90.83
FashionMNIST	RESNET50	32	Adam	0.0001	4	91.67	91.52
FashionMNIST	RESNET18	16	SGD	0.001	8	92.24	91.83
FashionMNIST	RESNET50	16	SGD	0.001	8	92.56	91.67
FashionMNIST	RESNET18	16	SGD	0.0001	8	90.36	89.71
FashionMNIST	RESNET50	16	SGD	0.0001	8	87.41	86.88
FashionMNIST	RESNET18	16	Adam	0.001	8	93.93	92.97
FashionMNIST	RESNET50	16	Adam	0.001	8	92.41	92.41
FashionMNIST	RESNET18	16	Adam	0.0001	8	93.6	92.7
FashionMNIST	RESNET50	16	Adam	0.0001	8	92.89	92.51
FashionMNIST	RESNET18	32	SGD	0.001	8	92.19	91.25
FashionMNIST	RESNET50	32	SGD	0.001	8	90.39	89.6
FashionMNIST	RESNET18	32	Adam	0.0001	8	93.16	92.53
FashionMNIST	RESNET50	32	Adam	0.0001	8	92.99	92.23



FashionMNIST - Training Curves



FashionMNIST - Model Performance Comparison



Q2 : Compute/time/FLOPs comparison

Single-epoch training runs for FashionMNIST were used to measure training time on CPU and GPU (BATCH_SIZE=16, NUM_EPOCHS=1, LR=0.001).

- Models: ResNet-18 and ResNet-50 (ResNet-32 optional).
- ptflops used to compute MACs and params. FLOPs per sample computed as $2 \times \text{MACs}$.
- Estimated epoch FLOPs computed as $\text{forward_flops_per_sample} \times \text{BATCH_SIZE} \times \text{steps} \times 3$.

Model	Device	Optimizer	LearningRate	BatchSize	Epochs	TrainTime	TrainAcc_pct	TestAcc_p	MACs_per_sampl	FLOPs_per_sampl	Params	Estimated_Epoch_FLOPs_total
resnet18	cpu	SGD	0.001	16	1	339685	79.1717	85.2	33201930	66403860	11175370	1.19527E+13
resnet18	cpu	Adam	0.001	16	1	392958	82.5367	87.27	33201930	66403860	11175370	1.19527E+13
resnet50	cpu	SGD	0.001	16	1	612882	64.7933	77.1	78726410	157452820	23522250	2.83415E+13
resnet50	cpu	Adam	0.001	16	1	723654	74.4367	81.74	78726410	157452820	23522250	2.83415E+13
resnet18	cuda	SGD	0.001	16	1	32334	79.19	85.32	33201930	66403860	11175370	1.19527E+13
resnet18	cuda	Adam	0.001	16	1	33521	82.7167	86.6	33201930	66403860	11175370	1.19527E+13
resnet50	cuda	SGD	0.001	16	1	79184	63.6983	77.48	78726410	157452820	23522250	2.83415E+13
resnet50	cuda	Adam	0.001	16	1	86606	75.6767	82.21	78726410	157452820	23522250	2.83415E+13

For a single training epoch on FashionMNIST (batch size 16, LR=0.001): ResNet-18 trains far faster than ResNet-50 and in these 1-epoch runs achieves higher test accuracy. Running on GPU provided large wall-clock speedups compared to CPU (≈ 7.7 – $11.7\times$ depending on model/optimizer). Adam gave consistently better test accuracy than SGD in our runs but at the cost of somewhat higher training time. ResNet-50 requires $\approx 2.37\times$ more FLOPs and $\approx 2.11\times$ more parameters than ResNet-18; however, the accuracy gains do not justify that compute in this 1-epoch setting, suggesting deeper models need more epochs / tuning.

GPU yields a large reduction in wall-clock time (roughly 8–12 $\times$). The smaller ResNet-18 shows slightly larger speedup ratios (≈ 10 – $12\times$) than ResNet-50 (≈ 7.7 – $8.4\times$) in these runs GPU time scales roughly with model FLOPs but CPU times are affected by other overheads (I/O, data loading, BLAS behavior), producing different CPU/GPU ratios across models.